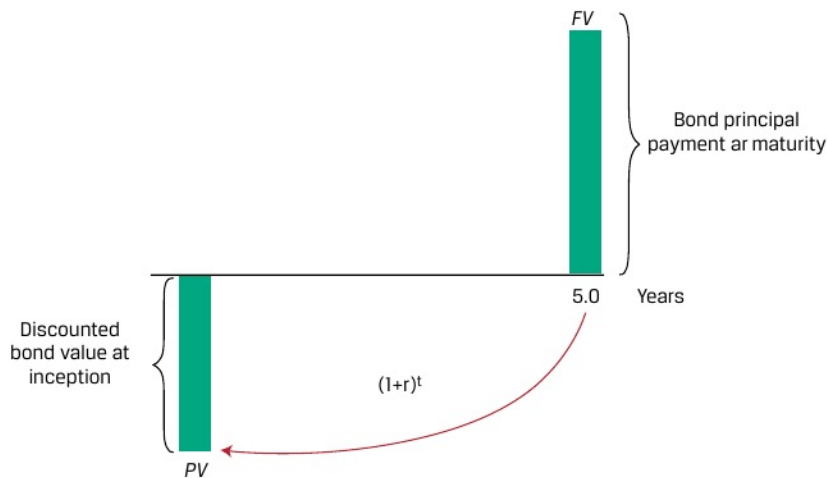


Exhibit 7: Discount Bond Implied Return



We may rearrange Equation 5 from to solve for the implied periodic return earned over the life of the instrument (t periods):

$$r = \sqrt[t]{\frac{FV_t}{PV}} - 1 = \left(\frac{FV_t}{PV}\right)^{\frac{1}{t}} - 1. \quad (18)$$

EXAMPLE 8

Discount Bond Implied Return

Recall from Example 1 that in 2016, German 10-year government bond investors faced a price of EUR100.50 per EUR100 principal and an annual YTM of -0.05 percent at issuance. That is, the German 10-year government bond was initially priced by the market to provide a negative return to the investor. Six years later, these bonds traded at a price (PV) of EUR95.72 per EUR100 principal.

1. What was the initial investor's implied return on this bond over the six-year holding period?

Solution:

-0.81 percent

We can solve for an investor's annualized return (r) using Equation 18 and a PV of 100.5, FV of 95.72, and t of 6 as follows:

$$r = -0.81\% = \left(\frac{95.72}{100.5}\right)^{\frac{1}{6}} - 1.$$

Note that an investor who purchases the discount bond at issuance and receives EUR100 in 10 years expects an implied return equal to the issuance YTM of -0.05 percent. However, the EUR4.78 price decline for the first six years translates into an annualized return of -0.81 percent, which is below the initial YTM of -0.05 percent. This negative return is consistent with an expected decline in the price (PV) of a discount bond amid higher inflation.